

SIMRET ABEBE

FinTech ML Engineer & Backend Developer

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SUMMARY

Computer Science student and backend developer with a strong focus on machine learning for financial systems. Skilled in building secure, scalable backend APIs with Django, and in designing ML pipelines that turn raw transaction, review, and complaint data into credit risk, fraud, and NLP models with measurable accuracy. Comfortable working across the full ML lifecycle from data cleaning and model training to deployment and production-grade authentication with a particular interest in solving financial inclusion problems for both the Ethiopian and international fintech markets.

PROFESSIONAL EXPERIENCE

AI & Machine Learning Intern, 10 Academy (Remote Internship)

Apr 2026 - August 2026 | Addis Ababa

- Built 3+ end-to-end AI and machine learning solutions for real-world Ethiopian fintech challenges, applying Python, Pandas, and scikit-learn across forecasting, credit risk modeling, and financial NLP workflows.
- Worked across the full ML lifecycle — data cleaning, feature engineering, model training, evaluation, and explainability on projects using real Ethiopian banking and financial complaint data.

Back End Developer, Melos Technologies PLC (Internship)

Jun 2026 - Jul 2026 | Addis Ababa

- Designed and implemented a JWT-based authentication system supporting Buyer, Seller, and Admin roles, including token refresh handling and route-level access control.
- Developed custom Django REST Framework APIs with role-based permissions, secure user management, and clean endpoint architecture built for production use and future scaling.

Web Development Intern, Prodigy InfoTech (Remote)

Apr 2025 – May 2025

- Built and delivered 4 responsive web applications using HTML5, CSS3, and vanilla JavaScript within a 2-month internship, focusing on cross-device consistency and clean UI structure.

KEY PROJECTS — 10 ACADEMY AI & ML INTERNSHIP

Ethiopian Bank Customer Review Intelligence

- Multilingual (English, Amharic, Afaan Oromo) ML system predicting customer risk from CBE, BOA, and Dashen Bank app reviews — achieved 100% recall on high-risk reviews.
- Delivered SHAP-based explainability and a live interactive dashboard for non-technical stakeholders to explore model output.

Credit Risk Probability Model

- Built an end-to-end ML pipeline for credit risk classification using alternative transaction data and RFM proxy variables, aligned with Basel II principles.
- Integrated SHAP for model interpretability and containerized the environment with Docker for reproducible, production-style deployment.

Fraud Detection System

- Compared Logistic Regression, Random Forest, XGBoost, and LightGBM on e-commerce and credit card transaction data; identified and fixed a cross-validation data leakage bug.
- Delivered SHAP-driven business recommendations, including risk scoring rules for new-account and geography-based fraud signals.

RAG-Powered Customer Complaint Chatbot

- Built a Retrieval-Augmented Generation conversational agent that analyzes financial customer complaints and retrieves evidence-backed answers via semantic search, reducing response hallucinations.

EDUCATION

B.Sc. in Computer Science, Bahir Dar University (BIT)

Expected Jul 2027

SKILLS

Languages: Python, JavaScript, SQL, HTML5, CSS3

Frameworks: Django, Django REST Framework, React.js, Next.js, Tailwind CSS, Node.js

Machine Learning & AI: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP (spaCy, DistilBERT), Sentiment Analysis, Statistical Modeling (ANOVA), SHAP Model Interpretability, XGBoost, LightGBM, scikit-learn

Data & Infrastructure: Docker, PostgreSQL, MySQL, MongoDB, Pandas, Vector Databases

Engineering Practices: Git & GitHub, REST APIs, JWT Authentication, CI/CD, Socket.io, Role-Based Access Control

VOLUNTEERING & ACHIEVEMENTS

G17 UAC Campus Ambassador — G17 Ethiopian Chapter, Nov 2025–Present. Represent Bahir Dar University on tech & AI innovation initiatives.

Hackathon Finalist — SmarHire AI, Bahir Dar Institute of Technology, Apr 2026. Built a full AI-powered platform in 48 hours.